456/1
MATHEMATICS
Paper 1
Oct./Nov.2026
$2\frac{1}{4}$ hours

UGANDA NATIONAL EXAMINATIONS BOARD

Uganda Certificate of Education

MATHEMATICS

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

This examination paper has **two** Sections; **A** and **B**. It consists of **six** items.

Section **A** has **two compulsory** items.

Section **B** has **two** parts; **I** and **II**. Respond to **one** item from **each** part.

Respond to **four examination items** in all.

Any additional item(s) responded to will **not** be scored.

All responses **must** be written in the answer booklet(s) provided.

Graph Paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

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SECTION A

*Respond to **both** items in this section.*

ITEM ONE

A manager of a company organised a sales promotion event in different market centres alongside a fundraising event that took place in the company. The sales promotion event had 27 female and 36 male workers. These were grouped into teams with each team having the same number of males and the same number of females.

The number of market centres visited were equal to the number of teams formed. This promotion increased the company sales by 18% from Shs. 48,000,000. During the fundraising, the board of directors (BOD) contributed ten million five hundred thousand shillings. The company staff contributed an amount that was $1\frac{1}{2}$ of the amount contributed by the BOD and the sales promotion team contributed 0.04 of the amount obtained after the increased company sales.

The total amount contributed during the fundraising was later donated to specific community schools and charity organisations in the ratio 5: 7 respectively.

The BOD has requested the manager to generate information clearly showing the;

I. Greatest number of market centres visited during the promotion,
II. Number of female and male workers in each team,
III. Amount to which the sales increased,
IV. Amount donated to schools and charity organisations.

However, the manager has failed to generate the report and needs your help.

TASK

Provide the information needed by the BOD.

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ITEM TWO

Nalunga has a right-angled triangular garden whose base is 50 m and height is 30 m. She has been informed that she can double her harvest if she expands the garden to a total area of 2400 m^2 by increasing both the base and the height by the same number of metres without changing the shape of the garden. However, she does not know how to do it.

Nalunga has been buying 40 kg of seeds for planting in her garden from a supplier who has been charging her a total amount of Shs. 650,000 that included the cost of the seeds as well as a fixed transport cost for delivering the seeds.

With the expanded garden Nalunga will need 60 kg of seeds for planting. The supplier has informed her that she will be charged a total amount of Shs. 950,000 for both the cost of the seeds and the fixed transport cost for delivering the seeds.

Nalunga's neighbour has a total of Shs. 500,000 which she intends to spend on both the seeds and their transport for delivery but she is not sure of the number of kilogrammes of seeds the supplier will deliver.

TASK

(a) Determine the;
   (i) Number of metres by which Nalunga should increase each side of the garden.
   (ii) Size of the base and the height after the increase.

(b) (i) Form an equation Nalunga and her neighbour would use to determine the amount of money to pay the supplier upon delivery of any number of kilogrammes of seeds.
   (ii) Inform Nalunga's neighbour, the number of kilogrammes of seeds the supplier will deliver for the total amount of money she intends to spend.

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SECTION B

*This section has **two** parts; **I** and **II**.*

Part I

*Respond to **one** item from this part.*

ITEM THREE

Hassad and Hajarah are planning to buy scholastic materials from the same shop. Hassad will buy 6 books, 4 pencils and 3 rubbers while Hajarah will buy 8 books, 5 pencils and 4 rubbers.

The price of each book, pencil and rubber is Shs. 3500, Shs. 600 and Shs. 200 respectively. Because prices are expected to increase in the near future, their parent has asked them to arrange the information about the materials as well as their respective prices in rows and columns, so that they request for money to buy **double** each of the materials they need. However, they do not know how to do it.

At the shop there is a back-to-school promotion where after buying scholastic materials, a customer has a chance of winning a gift of either a pen or a ruler by randomly picking two pieces of paper, one at a time, without replacement. If a customer picks two papers with the same item written on them, the customer wins the item.

There are 6 papers with a word **pen** written on them and 5 papers with a word **ruler** written on them.

Hassad and Hajarah wish to determine the probability of winning either a pen or a ruler.

TASK

(a) Organise the information as requested for by the parent.

(b) Determine the amount of money each one should request from their parent.

(c) Inform Hassad and Hajarah their chance of winning either a pen or a ruler.

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ITEM FOUR

A school has organized three seminars in three different terms of the year; the Science seminar, Arts seminar and Mathematics seminar.

Out of the 90 students of senior four, 42 have paid for the Arts seminar while 39 have paid for the Mathematics seminar. 15 students have paid for both the Arts and Mathematics seminars, 12 have paid for both the Science and Mathematics seminars while 21 have paid for both the Science and Arts seminars. The number of students who have paid for the Science seminar only is equal to the number of students who have paid for both the Science and Arts seminars only. Some students have paid for all the seminars. The ratio of the number of students who have paid for all the seminars to the number of students who have not paid for any of the seminars is 3:4.

On the day of the Science seminar, the teacher of English language wants to teach the remaining students only if the probability that a student selected at random from the entire class did not pay for the science seminar, is atleast 0.5.

TASK

Advise the teacher of English language, based on calculations, whether or not to prepare to teach the remaining students.

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Part II

*Respond to **one** item from this part.*

ITEM FIVE

John, who lives in town A, is a Loans Officer from JK SACCO. He intends to move from home to town Z for a meeting and then proceed to town Y in the same direction to collect money from a client who took a loan from the SACCO. Figure **1** shows a sketch map of the location of towns A, Z and Y.

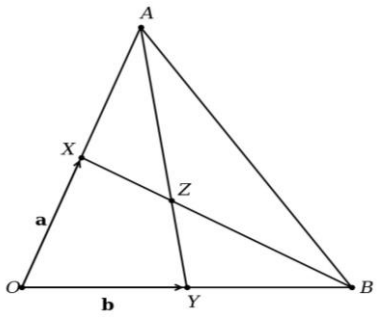


Fig. 1

X and Y are places in the middle of the routes OA and OB respectively. The routes XB and AY meet at Z. OA = 2**a**, OB = 2**b** and 3XZ = XB. AY is 60 km long.

John needs to use the sketch map and the information given to establish the vector AZ in terms of AY so as to determine the actual distance of AZ.

In addition, John has to be in town Z by 8:00 a.m. He is not sure whether he will make it in time if he leaves home at 7:30 a.m, driving at an average speed of 50 km/h. The client at Y borrowed Shs. 3,000,000 at a simple interest rate of 10% per annum for a period of two years. However, the client is doubtful of the total amount of money he should prepare for John to collect.

TASK

(a) Determine the:

(i) Relationship between vectors AZ and AY.

(ii) Actual distance of AZ in kilometres.

(b) Establish whether John will make it in time for the meeting at Z.

(c) Advise the client on the total amount of money he should prepare for John to collect.

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ITEM SIX

A monument in form of a pyramid in figure **2** is to be constructed with base PQR, in which PQ = PR = 16 m and QR = 20 m. The vertex, V, is 13 m vertically above P. The outer faces of the monument will be covered with bricks and it is known that 28 bricks cover 1 m^2.

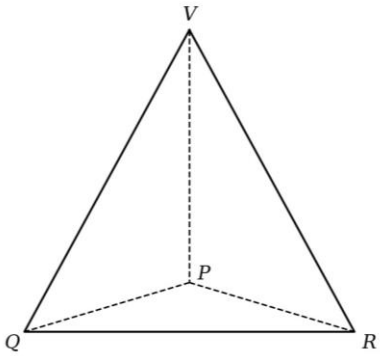


Fig. 2

A manager of a company that has been contracted to construct the monument intends to set off from his office and cycle due North at 15 km/h to move to the site of the monument. At the same time, the company supervisor also intends to set off from her home, 7.5 km due East of the Manager's office and cycle on a bearing of 340° so that she meets the manager before reaching the site of the monument. The supervisor however, does not know how fast she will have to cycle if she is to meet the manager.

The manager also needs to know the number of bricks that will be required for the outer face of the monument.

TASK

(a) Estimate, to the nearest 1000, the number of bricks that will be required.

(b) Recommend, based on calculations, the speed with which the supervisor will travel if she is to meet the manager.

END

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PROPOSED GUIDE
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SECTION A

ITEM ONE (NUMBERS)

Task part (i)

Greatest number of market centres visited during the promotion is equal to the Greatest Common Factor (GCF) of 27 and 36.

3	27	36
3	9	12
	3	4

GCF of 27 and 36 $= 3 \times 3$

$= 9$

Therefore, there were 9 teams formed.

Since the number of market centres visited is equal to the number of teams formed, then the greatest number of market centres visited during the promotion was 9.

Task part (ii)

Number of female workers in each team.

$$= \frac{\text{Total number of female workers available.}}{\text{Total number of teams created.}}$$

$$= \frac{27}{9}$$

$= 3$ Female workers per team.

Number of male workers in each team.

$$= \frac{\text{Total number of male workers available.}}{\text{Total number of teams created.}}$$

$$= \frac{36}{9}$$

$= 4$ Male workers per team.

Therefore, each team had 3 females and 4 males.

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Task part (iii)

Increase in the company sales

$= 18\%$ of Shs. 48,000,000

$= \frac{18}{100} \times$ Shs. 48,000,000

$=$ Shs. $(18 \times 480{,}000)$

$=$ Shs. 8,640,000

New amount of sales of the company

$=$ Old amount of sales $+$ Increase in sales

$=$ Shs. 48,000,000 + Shs. 8,640,000

$=$ Shs. 56,640,000

Therefore, the company sales increased to Shs. 56,640,000

Task part (iv)

Amount of money contributed by the BOD

	Shs. 10,000,000
+	Shs. 500,000
	Shs. 10,500,000

Amount of money contributed by the Company staff

$= 1\frac{1}{2}$ of Shs. 10,500,000

$= \frac{3}{2} \times$ Shs. 10,500,000

$=$ Shs. 15,750,000

Amount of money contributed by the Sales promotion team

$= 0.04$ of Shs. 56,640,000

$= 0.04 \times$ Shs. 56,640,000

$=$ Shs. 2,265,600

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Total amount of money contributed during the fund raising

	Shs. 10,500,000
+	Shs. 15,750,000
	Shs. 2,265,600
	Shs. 28,515,600

Total ratio

$= 5 + 7$

$= 12$ Parts

Amount of money donated to schools.

$= \frac{5}{12} \times$ Shs. 28,515,600

$=$ Shs. 11,881,500

Amount of money donated to schools.

$= \frac{7}{12} \times$ Shs. 28,515,600

$=$ Shs. 16,634,100

Therefore, eleven million eight hundred eighty-one thousand five hundred shillings was donated to schools and sixteen million six hundred thirty-four thousand one hundred shillings was donated to charity organizations.

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ITEM TWO (PATTERNS AND ALGEBRA)

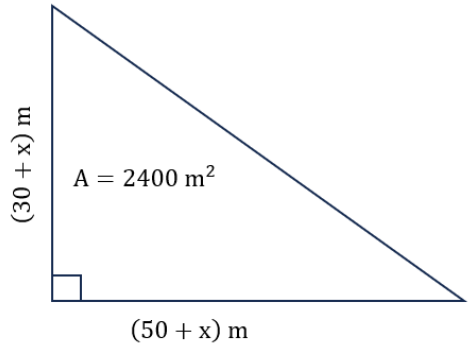
Task part (a)

i)

Let x represent the number of metres that Nalunga should add to the base and height of her garden.

Item	Old value	New value
Base	50 m	$(50 + x)$ m
Height	30 m	$(30 + x)$ m
Area		$A = 2400\ m^2$

Diagram showing Nalunga's garden after expansion.



Area of the triangle $= 2400\ m^2$

$\frac{1}{2}(50 + x)(30 + x) = 2400$

$(50 + x)(30 + x) = 2400 \times 2$

$50(30 + x) + x(30 + x) = 4800$

$1500 + 50x + 30x + x^2 = 4800$

$1500 + 80x + x^2 = 4800$

$x^2 + 80x + 1500 - 4800 = 0$

$x^2 + 80x - 3300 = 0$

Product $= (1 \times -3300)$

Product $= -3300$

Sum $= 80$

The factors of -3300 that give a sum of 80 are 110 and -30

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$x^2 + 110x - 30x - 3300 = 0$

$x(x + 110) - 30(x + 110) = 0$

$(x + 110)(x - 30) = 0$

Either $(x + 110) = 0$ or $(x - 30) = 0$

$x + 110 = 0$ $x - 30 = 0$

$x = -110$ m $x = 30$ m

Since distance (length of a garden) cannot be less than zero (negative), then the value of x is 30m

Therefore, Nalunga should increase each side of her garden by a distance of 30m

ii)

Size of the base after increase.

$= 50$ m $+ 30$ m

$= 80$ m

Size of the height after increase.

$= 30$ m $+ 30$ m

$= 60$ m

Therefore, after the increase, the base of Nalunga's garden will be of length 80 m and the height of the garden will be of length 60 m

Task part (b)

i)

Let x represent the number of Kilograms of seeds supplied and let y represent the total amount charged for supplying the seeds.

A table showing the summary of the given information

x (kg)	40	60
y (Shs)	650,000	950,000

Using the equation of a straight line,

$y = mx + c$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$m = \frac{950{,}000 - 650{,}000}{60 - 40}$

$m = \frac{300{,}000}{20}$

$m = 15{,}000$

But $y = mx + c$

$650{,}000 = 15{,}000(40) + c$

$650{,}000 - 15{,}000(40) = c$

$650{,}000 - 600{,}000 = c$

$c = 50{,}000$

The equation becomes $y = 15000x + 50{,}000$

The equation that Nalunga and her neighbour would use to determine the amount of money to pay the supplier (y) upon delivery of any number of kilogrammes of seeds (x) is $y = 15000x + 50{,}000$

ii)

Determining the value of x when $y = \text{Shs. } 500{,}000$

$500{,}000 = 15{,}000x + 50{,}000$

$500{,}000 - 50{,}000 = 15{,}000x$

$450{,}000 = 15{,}000x$

$\frac{450{,}000}{15{,}000} = \frac{15{,}000x}{15{,}000}$

$x = 30\text{kg}$

Therefore, the supplier will deliver 30kg of seeds for the Shs. 500,000 that Nalunga's neighbour intends to spend.

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SECTION B

Part I (DATA AND PROBABILITY)

ITEM THREE

A table showing the summary of the given information

	Books	Pencils	Rubbers
Hassad	6	4	3
Hajarah	8	5	4
Unit Cost	Shs. 3,500	Shs. 600	Shs. 200

Task part (a)

A matrix showing the scholastic materials that Hassad and Hajarah are planning to buy.

$$\begin{pmatrix} 6 & 4 & 3 \\ 8 & 5 & 4 \end{pmatrix}$$

A matrix showing the unit cost of books, pencils and rubbers

$$\begin{pmatrix} \text{Shs. } 3{,}500 \\ \text{Shs. } 600 \\ \text{Shs. } 200 \end{pmatrix}$$

Task part (b)

A matrix showing the scholastic materials that Hassad and Hajarah should request for.

$$= 2\begin{pmatrix} 6 & 4 & 3 \\ 8 & 5 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 12 & 8 & 6 \\ 16 & 10 & 8 \end{pmatrix}$$

Amount of money each student should ask for

$$= \begin{pmatrix} 12 & 8 & 6 \\ 16 & 10 & 8 \end{pmatrix} \times \begin{pmatrix} \text{Shs. } 3{,}500 \\ \text{Shs. } 600 \\ \text{Shs. } 200 \end{pmatrix}$$

$$= \begin{pmatrix} (12 \times \text{Shs. } 3{,}500) + (8 \times \text{Shs. } 600) + (6 \times \text{Shs. } 200) \\ (16 \times \text{Shs. } 3{,}500) + (10 \times \text{Shs. } 600) + (8 \times \text{Shs. } 200) \end{pmatrix}$$

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$$= \begin{pmatrix} \text{Shs. } 42{,}000 + \text{Shs. } 4{,}800 + \text{Shs. } 1{,}200 \\ \text{Shs. } 56{,}000 + \text{Shs. } 6{,}000 + \text{Shs. } 1{,}600 \end{pmatrix}$$

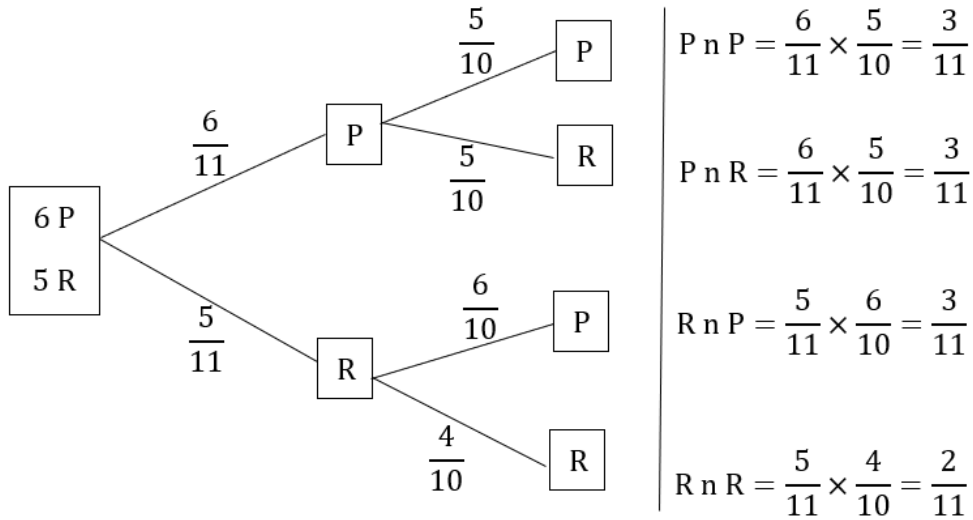
$$= \begin{pmatrix} \text{Shs. } 48{,}000 \\ \text{Shs. } 63{,}600 \end{pmatrix}$$

Therefore, Hassad should request for Shs. 48,000 and Hajarah should request for Shs. 63,600 from their parent.

Task part (c)

Let P represent the event of picking a pent and let R represent the event of picking a ruler.

A probability tree diagram showing the sample space for the experiment.



Chance of winning an item from the shop.

= Probability of picking two papers with the same item written on it.

$= P(PnP) + P(RnR)$

$$= \frac{3}{11} + \frac{2}{11}$$

$$= \frac{5}{11}$$

Therefore, the chance of Hassad and Hajarah winning either a pen or ruler from the shop is $\frac{5}{11}$.

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ITEM FOUR (GEOMETRY AND MEASURES)

Let

S represent the event of paying for the Science seminar,

A represent the event of paying for the Arts seminar,

M represent the event of paying for the Mathematics seminar,

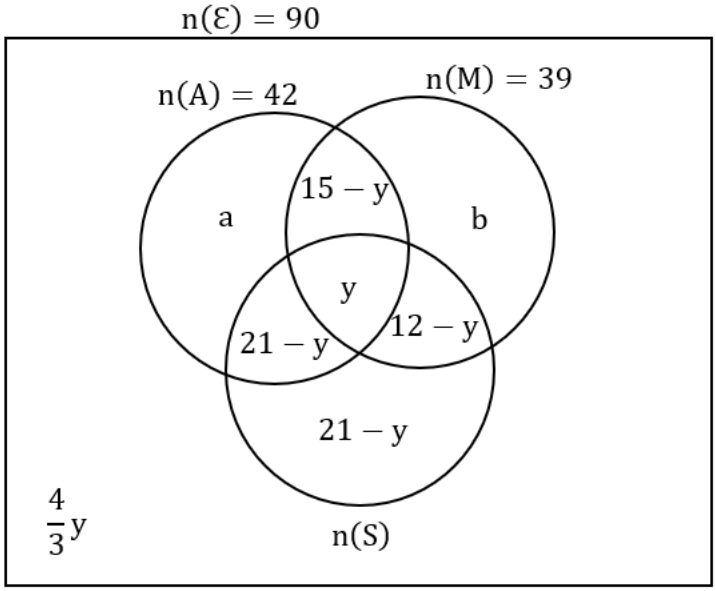
y represent the number of students who paid for all the three seminars,

a represent the number of students who paid for the Arts seminar only,

b represent the number of students who paid for the Mathematics seminar only,

c represent the number of students who paid for the Science seminar only

A Venn diagram showing the number of students who paid for Science, Arts and Mathematics seminars in a school



From the Venn diagram,

$a + 15 - y + y + 21 - y = 42$

$a + 15 + 21 - y = 42$

$a + 36 - y = 42$

$a - y = 42 - 36$

$a - y = 6$

$a = 6 + y$ ---------- equation (i)

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Similarly,

$b + 15 - y + y + 12 - y = 39$

$b + 15 + 12 - y = 39$

$b + 27 - y = 39$

$b - y = 39 - 27$

$b - y = 12$

$b = 12 + y$ ---------- equation (ii)

Also,

$a + b + 15 - y + y + 21 - y + 12 - y + \frac{4}{3}y + 21 - y = 90$

$a + b + 15 + 21 + 12 + 21 - 3y + \frac{4}{3}y = 90$

$a + b + 69 - 3y + \frac{4}{3}y = 90$

$a + b - 3y + \frac{4}{3}y = 90 - 69$

$a + b - \frac{5}{3}y = 21$ ---------- equation (iii)

Substituting equation (i) and equation (ii) into equation (iii)

$a + b - \frac{5}{3}y = 21$

$6 + y + 12 + y - \frac{5}{3}y = 21$

$6 + 12 + 2y - \frac{5}{3}y = 21$

$18 + \frac{1}{3}y = 21$

$\frac{1}{3}y = 21 - 18$

$\frac{1}{3}y = 3$

$\frac{y}{3} = 3$

$y = 3 \times 3$

$y = 9$

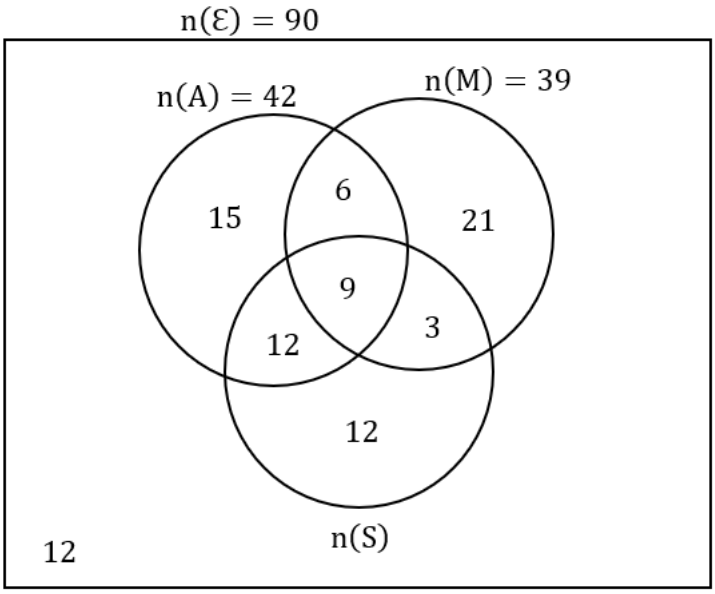
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Therefore,

$a = 6 + 9 = 15$

$b = 12 + 9 = 21$

A Venn diagram showing the number of students who paid for Science, Arts and Mathematics seminars in a school



Number of students who did not pay for the science seminar

$= 12 + 15 + 6 + 21$

$= 54$

Probability that a student selected from the entire class did not pay for the Science seminar.

$$= \frac{\text{Number of students who did not pay for the science seminar}}{\text{Total number of students in the entire class}}$$

$$= \frac{54}{90}$$

$= 0.6$ (which is greater than 0.5)

Therefore, since the probability that a student selected from the entire class did not pay for the Science seminar is greater than 0.5, the teacher of English language should prepare to teach the remaining students.

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ITEM FIVE

Task part (a)

i)

From the diagram,

$\overline{\text{AZ}} = \overline{\text{AX}} + \overline{\text{XZ}}$

$\overline{\text{AZ}} = \overline{-\text{XA}} + \overline{\text{XZ}}$

$\overline{\text{AZ}} = -\frac{1}{2}\overline{\text{OA}} + \frac{\overline{\text{XB}}}{3}$

$\overline{\text{AZ}} = -\frac{1}{2}\overline{\text{OA}} + \frac{\overline{\text{XB}}}{3}$

$\overline{\text{AZ}} = -\frac{1}{2}(2\boldsymbol{a}) + \frac{1}{3}(\overline{\text{XO}} + \overline{\text{OB}})$

$\overline{\text{AZ}} = -\boldsymbol{a} + \frac{1}{3}\left(-\frac{1}{2}\overline{\text{OA}} + \overline{\text{OB}}\right)$

$\overline{\text{AZ}} = -\boldsymbol{a} + \frac{1}{3}\left(-\frac{1}{2}(2\boldsymbol{a}) + 2\boldsymbol{b}\right)$

$\overline{\text{AZ}} = -\boldsymbol{a} + \frac{1}{3}(-\boldsymbol{a} + 2\boldsymbol{b})$

$\overline{\text{AZ}} = -\boldsymbol{a} - \frac{1}{3}\boldsymbol{a} + \frac{2}{3}\boldsymbol{b}$

$\overline{\text{AZ}} = -\frac{4}{3}\boldsymbol{a} + \frac{2}{3}\boldsymbol{b}$

$\overline{\text{AZ}} = \frac{2}{3}\boldsymbol{b} - \frac{4}{3}\boldsymbol{a}$

$\overline{\text{AZ}} = \frac{2\boldsymbol{b} - 4\boldsymbol{a}}{3}$

$\overline{\text{AZ}} = \frac{2(\boldsymbol{b} - 2\boldsymbol{a})}{3}$

$\overline{\text{AZ}} = \frac{2}{3}(\boldsymbol{b} - 2\boldsymbol{a})$

Similarly,

$\overline{\text{AY}} = \overline{\text{AO}} + \overline{\text{OY}}$

$\overline{\text{AY}} = -\overline{\text{OA}} + \overline{\text{OY}}$

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$\overline{\text{AY}} = -2\boldsymbol{a} + \frac{1}{2}\overline{\text{OB}}$

$\overline{\text{AY}} = -2\boldsymbol{a} + \frac{1}{2}(2\boldsymbol{b})$

$\overline{\text{AY}} = -2\boldsymbol{a} + \boldsymbol{b}$

$\overline{\text{AY}} = \boldsymbol{b} - \boldsymbol{2a}$

Ratio of $\overline{\text{AZ}}$ to $\overline{\text{AY}}$

$\overline{\text{AZ}} : \overline{\text{AY}}$

$= \frac{\overline{\text{AZ}}}{\overline{\text{AY}}}$

$\frac{\overline{\text{AZ}}}{\overline{\text{AY}}} = \frac{2}{3}(\boldsymbol{b} - 2\boldsymbol{a}) \div (\boldsymbol{b} - \boldsymbol{2a})$

$\frac{\overline{\text{AZ}}}{\overline{\text{AY}}} = \frac{2(\boldsymbol{b} - 2\boldsymbol{a})}{3} \times \frac{1}{(\boldsymbol{b} - 2\boldsymbol{a})}$

$\frac{\overline{\text{AZ}}}{\overline{\text{AY}}} = \frac{2}{3}$

$\overline{\text{AZ}} = \frac{2}{3}\overline{\text{AY}}$

Therefore, the relationship between vectors AZ and AY is that vector AZ is equal to two thirds of vector AY

ii)

Finding the actual distance of AZ in kilometers.

From $\overline{\text{AZ}} = \frac{2}{3}\overline{\text{AY}}$

$\overline{\text{AZ}} = \frac{2}{3}(60 \text{ km})$

$\overline{\text{AZ}} = 2 \times 20 \text{ km}$

$\overline{\text{AZ}} = 40 \text{ km}$

Therefore, the actual distance of AZ is 40 km

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Task part (b)

Let t represent the time that John spends driving from A to Z at 50 km/h

Using,

$$\text{Speed} = \frac{\text{Total distance}}{\text{Time taken}}$$

$$50 = \frac{40}{t}$$

$$50t = 40$$

$$t = \frac{40}{50}$$

$$t = \frac{4}{5} \text{ Hours}$$

John will take $\frac{4}{5}$ hours to drive from A to Z

Converting $\frac{4}{5}$ hours to minutes.

$$\frac{4}{5} \text{ Hours} = \frac{4}{5} \times 60$$

$$= 4 \times 12$$

$$= 48 \text{ minutes}$$

Arrival time = Departure time + duration for the jurney.

Arrival time = 07: 30 a. m + 00: 48.

Arrival time = 08: 18 a. m

John will arrive in town Z at 8: 18 a. m, meaning he will be late for meeting by 18 minutes.

Therefore, John will not make it in time for the meeting at Z

Task part (c)

Using the simple Interest Formula,

$$I = PTR$$

$$I = \text{Shs. } 3{,}000{,}000 \times 2 \times \frac{10}{100}$$

$$I = \text{Shs. } 3{,}000{,}000 \times 0.2$$

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I = Shs. 600,000

Total Amount to be prepared for John to collect

A = Principal + Interest

A = Shs. 3,000,000 + Shs. 600,000

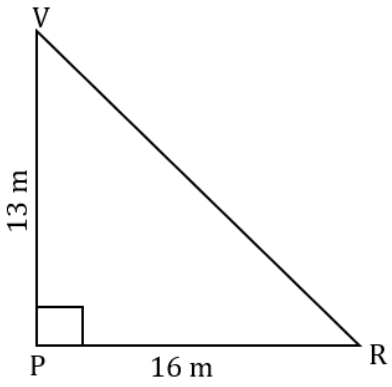
A = Shs. 3,600,000

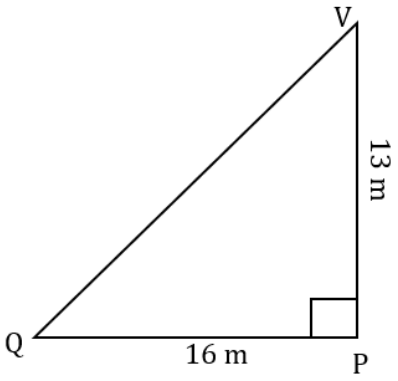
Therefore, the client should prepare a total of Shs. 3, 600, 000 for John to collect.

ITEM SIX

Task part (a)

Considering Triangle PVR and Triangle PVQ





Area of Triangle PVR

$A = \frac{1}{2} \times b \times h$

$A = \frac{1}{2} \times 16\ m \times 13\ m$

$A = 8\ m \times 13\ m$

$A = 104\ m^2$

Area of Triangle PVQ

$A = \frac{1}{2} \times b \times h$

$A = \frac{1}{2} \times 16\ m \times 13\ m$

$A = 8\ m \times 13\ m$

$A = 104\ m^2$

Using the Pythagoras theorem to determine the lengths VR and VQ

$\overline{VP}^2 + \overline{PR}^2 = \overline{VR}^2$

$13^2 + 16^2 = \overline{VR}^2$

$169 + 256 = \overline{VR}^2$

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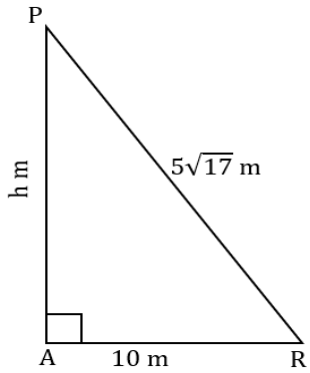
$425 = \overline{VR}^2$

$\sqrt{425} = \overline{VR}$

$\overline{VR} = 5\sqrt{17}$ m

Considering Triangle PQR

Let Point A be the midpoint of line QR and let h represent the length AP



Finding the value of h using the Pythagoras theorem.

$\overline{AP}^2 + \overline{AR}^2 = \overline{PR}^2$

$h^2 + 10^2 = \left(5\sqrt{17}\right)^2$

$h^2 + 100 = (25 \times 17)$

$h^2 + 100 = 425$

$h^2 = 425 - 100$

$h = \sqrt{325}$

$h = 5\sqrt{13}$ m

Area of triangle PQR

$A = \frac{1}{2} \times b \times h$

$A = \frac{1}{2} \times 20 \times 5\sqrt{13}$

$A = 10 \times 5\sqrt{13}$

$A = 50\sqrt{13}$ m^2

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Total surface area to be covered by the bricks.

= Area of Triangle PVR + Area of triangle PQR + Area of Triangle PVQ

$= 104\ m^2 + 50\sqrt{13}\ m^2 + 104\ m^2$

$= 208\ m^2 + 50\sqrt{13}\ m^2$

$= 388.27756\ m^2$

Number of bricks that will be required

$= 28 \times 388.27756$

= 10,871.77168 Bricks

Rounding off to the nearest 1000

10,871.77168 Bricks ≈ 11,000 Bricks.

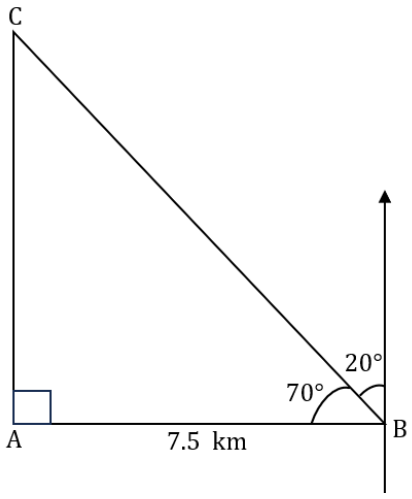
Therefore, about 11,000 bricks will be required for the outer face of the monument.

Task part (b)

Let point A represent the Manager's office, point B represent the Supervisor's home and point C represent the site of the monument.

Let t represent the time taken by the manager to move from his home to the site.

An illustration Showing the Journies of the manager and the supervisor.



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Applying trigonometry on triangle ABC to determine distance AC and BC.

$\tan 70° = \frac{AC}{AB}$ $\qquad$ $\cos 70° = \frac{AB}{BC}$

$\tan 70° = \frac{AC}{7.5\text{ km}}$ $\qquad$ $\cos 70° = \frac{7.5}{BC}$

$7.5\tan 70° = AC$ $\qquad$ $BC = \frac{7.5}{\cos 70°}$

$AC = 20.60608$ km $\qquad$ $BC = 21.92853$ km

The distance from the manager's office to the site of the monument is 20.60608 km.

From Speed $= \frac{\text{Distance}}{\text{time}}$

$15 = \frac{20.60608}{t}$

$15t = 20.60608$

$t = \frac{20.60608}{15}$

$t = 1.37374$ Hours.

The manager needs 1.37374 hours to move from his office to the site of the monument.

Speed of the supervisor

From Speed $= \frac{\text{Distance}}{\text{time}}$

$\text{Speed} = \frac{21.92853}{1.37374}$

$\text{Speed} = 15.96265$ km/h

I recommend the supervisor to move at a speed higher than 15.96265 km/h so as to meet the manager before reaching the site of the monument.

END

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